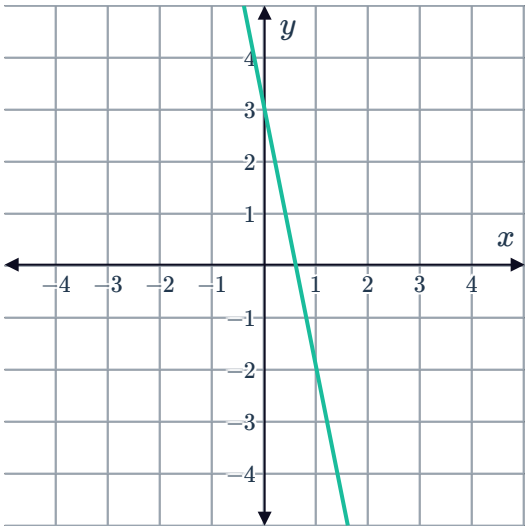
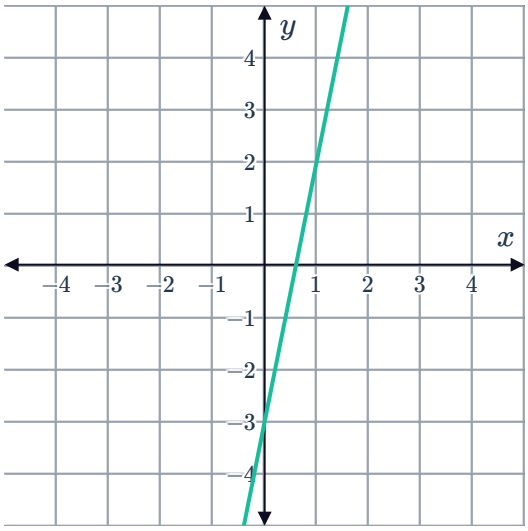
Increasing and decreasing functions

1 State whether each of the following linear functions are increasing or decreasing:

a



b

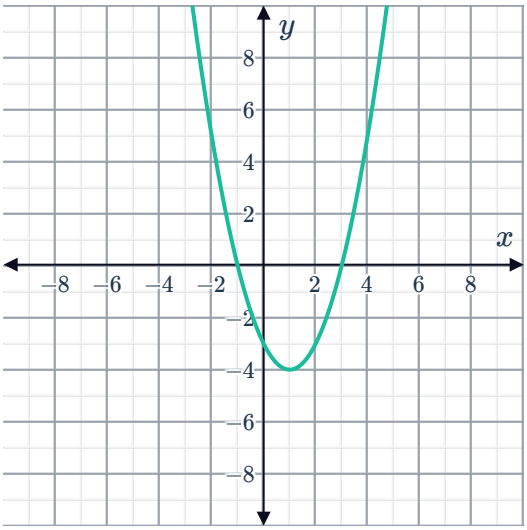


2 Consider the graph of the function $f(x) = (x+1)(x-3)$:

a State the x-value of the point where $f(x)$ changes from decreasing to increasing.

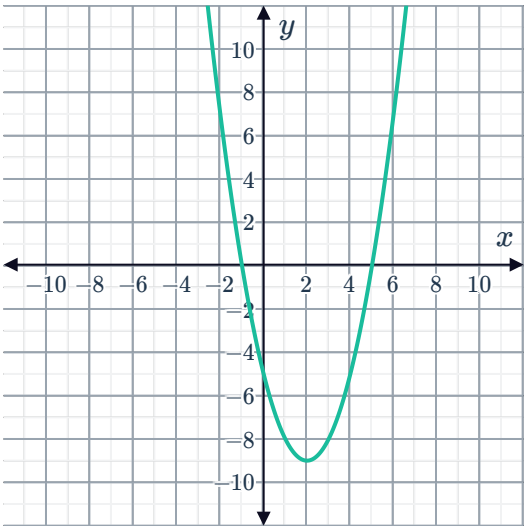
b State the interval where $f(x)$ is increasing.

c State the interval where $f(x)$ is decreasing.



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3 Consider the given graph. State the interval where $f(x)$ is decreasing.

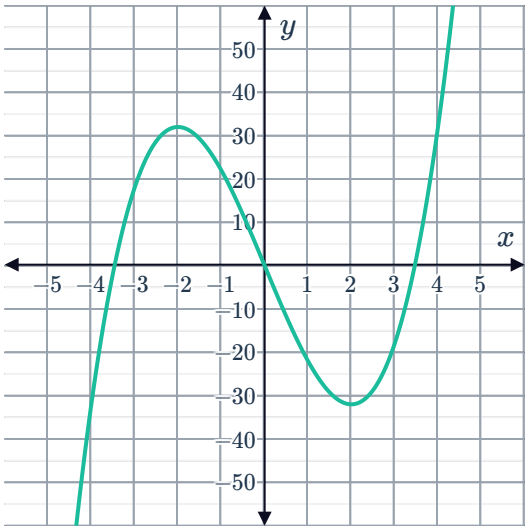


4 For each of the following functions:

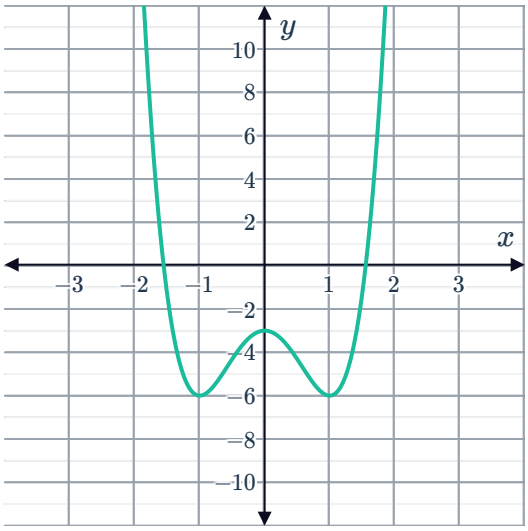
i State the interval on which the function is increasing.

ii State the interval on which the function is decreasing.

a $y = 2x^3 - 24x$

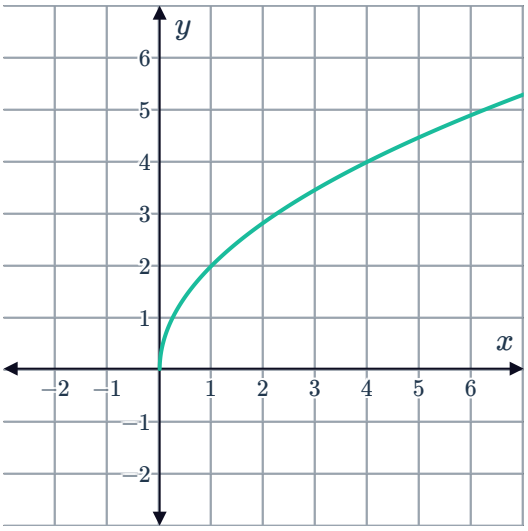


b $y = 3x^4 - 6x^2 - 3$



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5 Consider the graph:

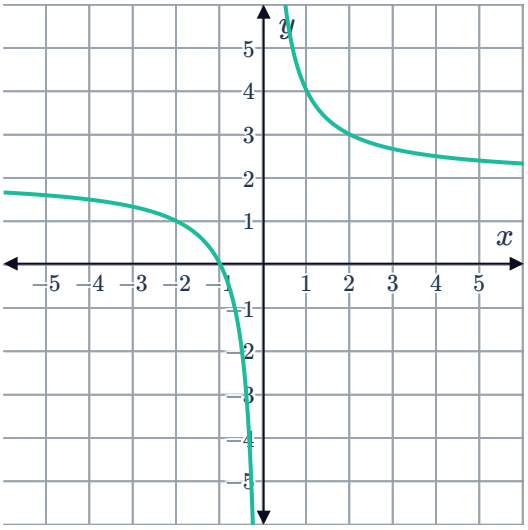


a State the coordinates of the x-intercept.

b State the coordinates of the y-intercept.

c Is the function increasing or decreasing as $x \to \infty$?

6 Consider the given graph:



a State the coordinates of the x-intercept.

b As $x \to +\infty$, does y increase or decrease?

c As $x \to -\infty$, does y increase or decrease?

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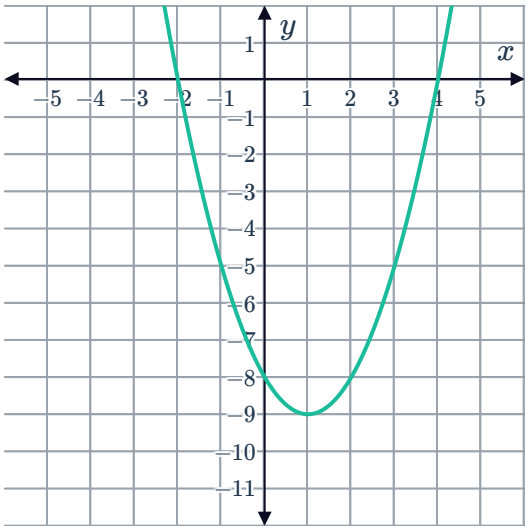
Zeros of functions

7 For each of the following graphs:

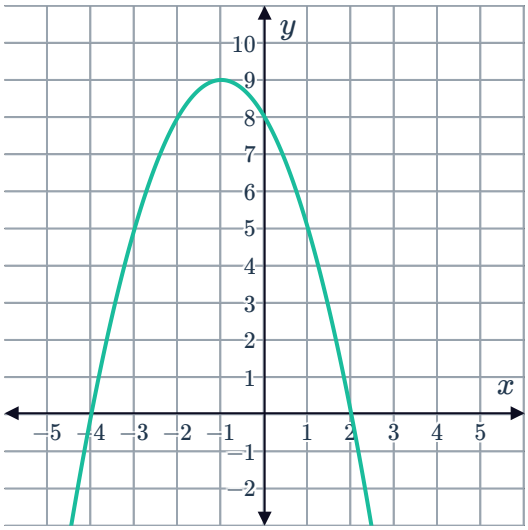
i Find the coordinates of the x-intercept(s).

ii Find the zeros of the function.

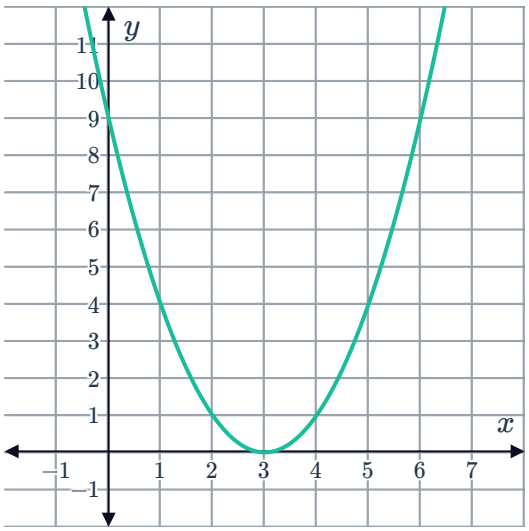
a



b



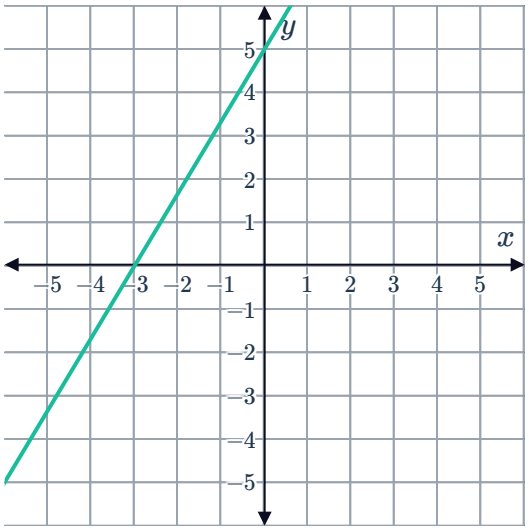
c



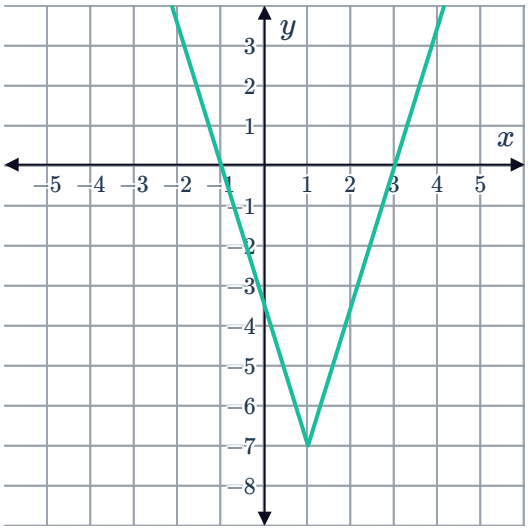
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8 Find the zeros of the following functions:

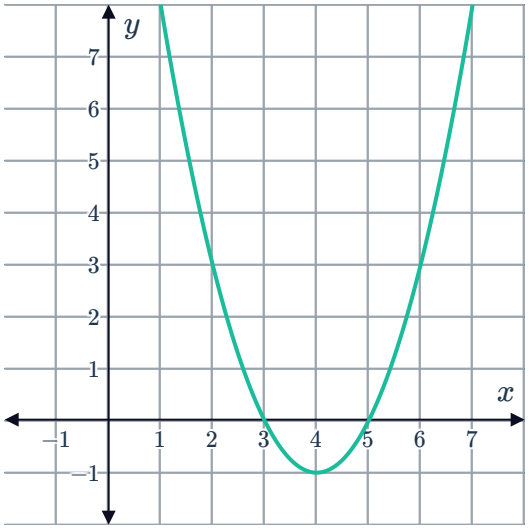
a



b



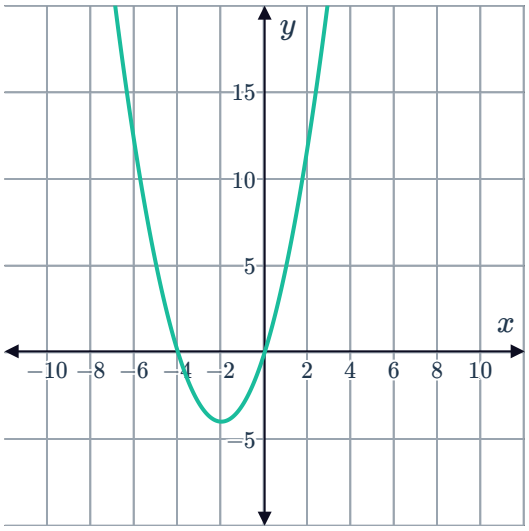
c



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End behavior

9 Consider the given graph:



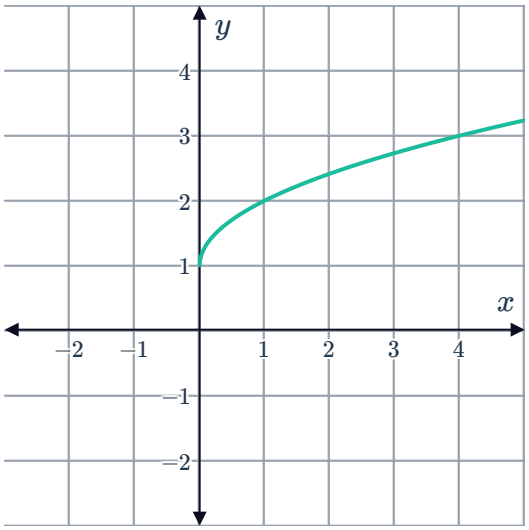
a State the coordinates of the x-intercepts.

b State the coordinates of the y-intercept.

c As $x \to +\infty$, what does y approach?

d As $x \to -\infty$, what does y approach?

10 Consider the given graph:



a State the coordinates of the x-intercepts.

b State the coordinates of the y-intercept.

c As $x \to +\infty$, what does y approach?

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11 For each of the following graphs:

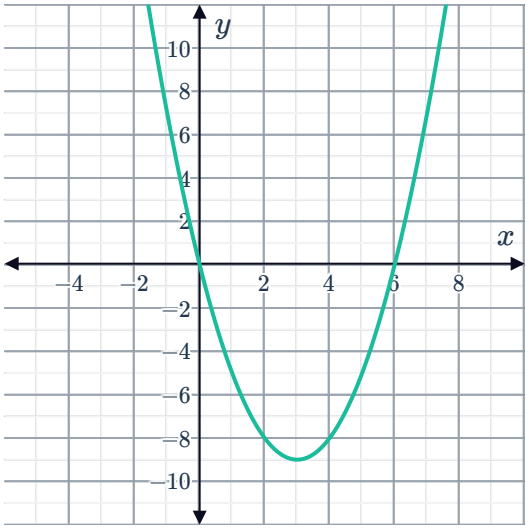
i State the coordinates of the x-intercepts.

ii State the coordinates of the y-intercepts.

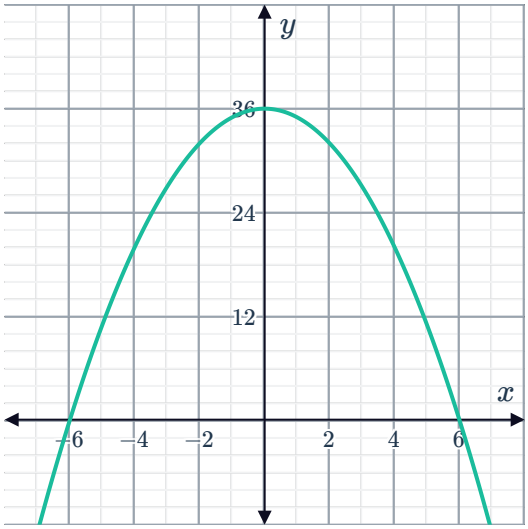
iii Describe the behavior of the function as $x \to \infty$.

iv Describe the behavior of the function as $x \to -\infty$.

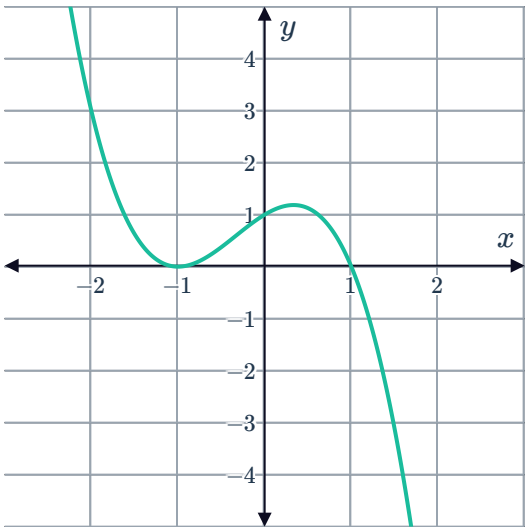
a



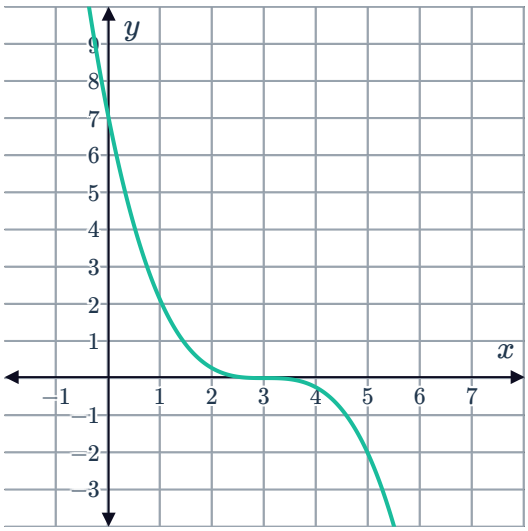
b



c

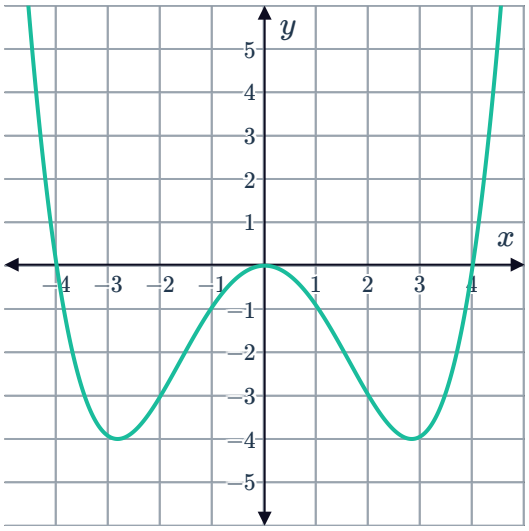


d

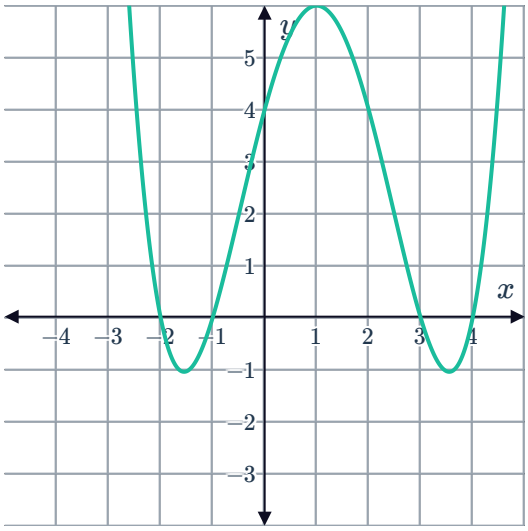


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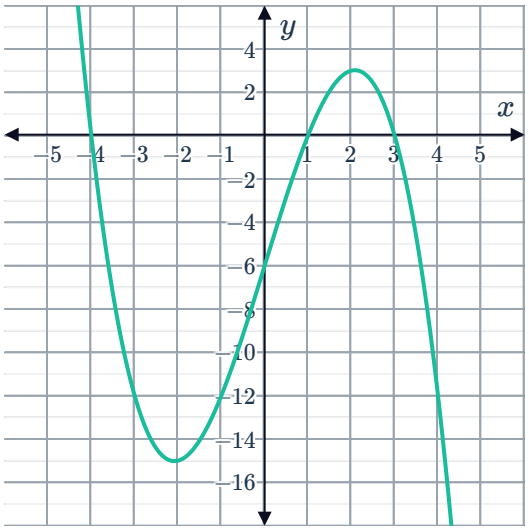
e



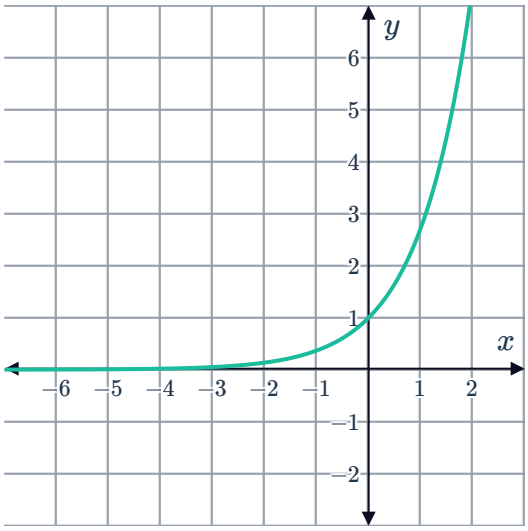
f



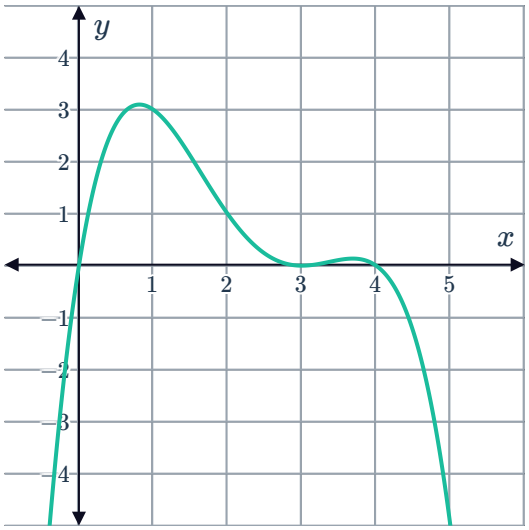
g



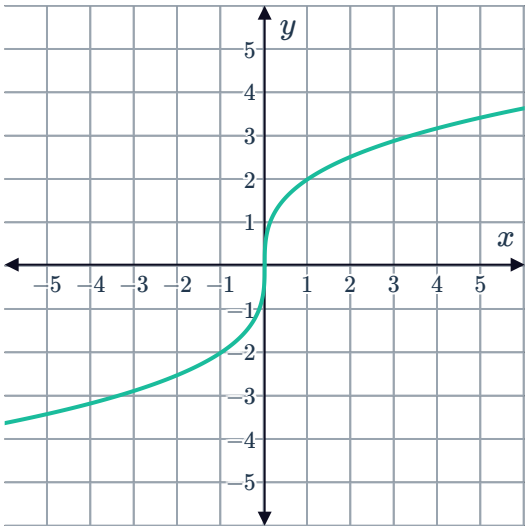
h



i



j



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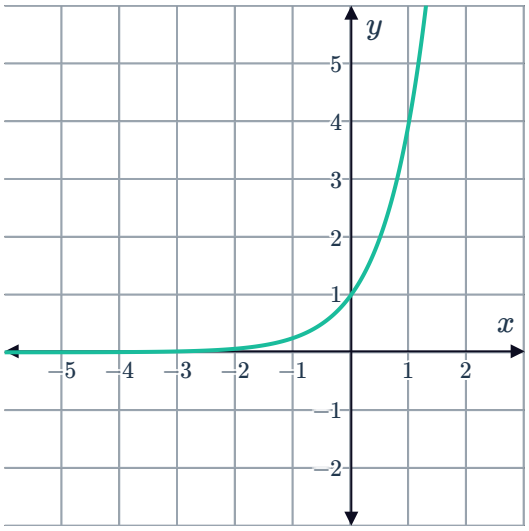
Asymptotes

12 A graph of the function $f(x) = 4^x$ is shown below:

a As $x \to -\infty$, what does the function approach?

b State the equation of the horizontal asymptote of $f(x)$

c True or false: The function is continuous over the interval of all real numbers, $(-\infty, \infty)$.

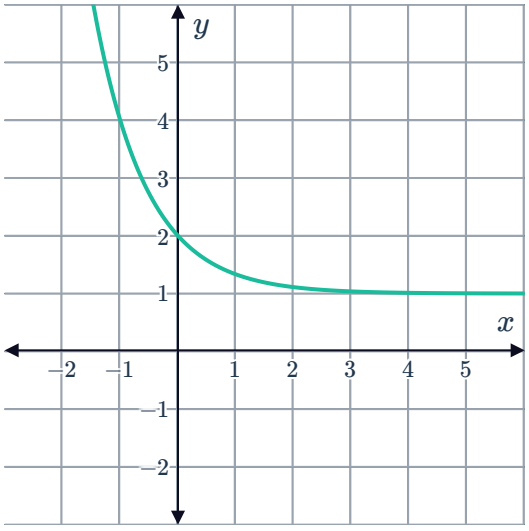


13 A graph of the function $f(x) = 3^{-x} + 1$ is shown below:

a As $x \to \infty$, what does the function approach?

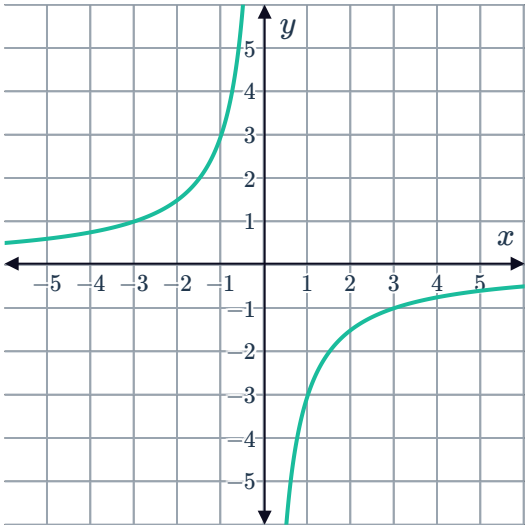
b State the equation of the horizontal asymptote of $f(x)$

c True or false: The function is continuous over the interval of all real numbers, $(-\infty, \infty)$.



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14 A graph of the function $f(x) = -\dfrac{3}{x}$ is shown below:



a Complete the following statements:

i If x is positive, what value does the function approach as x approaches 0?

ii If x is negative, what value does the function approach as x approaches 0?

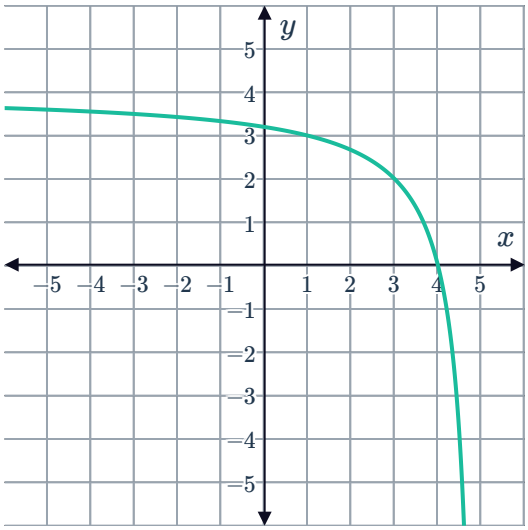
iii As $x \to \infty$, what does the function approach?

iv As $x \to -\infty$, what does the function approach?

b State the equation of the vertical asymptote of $f(x)$.

c State the equation of the horizontal asymptote of $f(x)$.

15 A graph of the function $f(x) = 4 + \dfrac{4}{x-5}$ is shown below:



a Complete the following statements:

i If $x > 5$, what value does the function approach as x approaches 5?

ii If $x < 5$, what value does the function approach as x approaches 5?

iii As $x \to \infty$, what does the function approach?

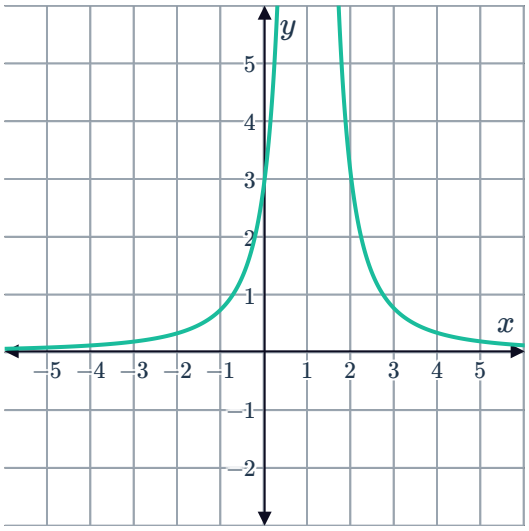
iv As $x \to -\infty$, what does the function approach?

b State the equation of the vertical asymptote of $f(x)$.

c State the equation of the horizontal asymptote of $f(x)$.

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16 A graph of the function $f(x) = \dfrac{3}{(x-1)^2}$ is shown below:



a Complete the following statements:

i If $x > 1$, what value does the function approach as x approaches 1?

ii If $x < 1$, what value does the function approach as x approaches 1?

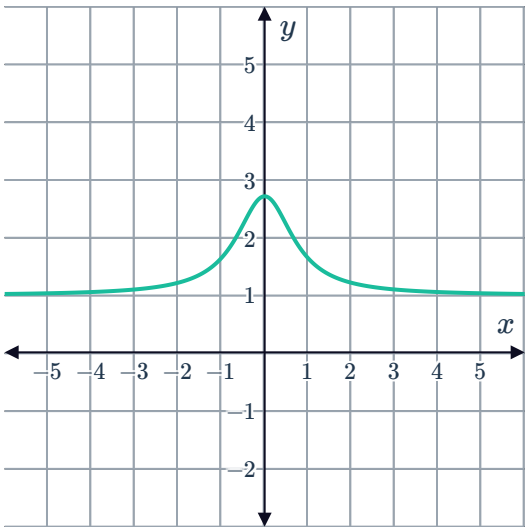
iii As $x \to \infty$, what does the function approach?

iv As $x \to -\infty$, what does the function approach?

b State the equation of the vertical asymptote of $f(x)$.

c State the equation of the horizontal asymptote of $f(x)$.

17 The graph of a function $f(x)$ is shown below:



a Complete the following statements:

i As $x \to \infty$, what does the function approach?

ii As $x \to -\infty$, what does the function approach?

b State the equation of the horizontal asymptote.

c State the largest interval of x-values that includes $x = 0$ over which the function is continuous.

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